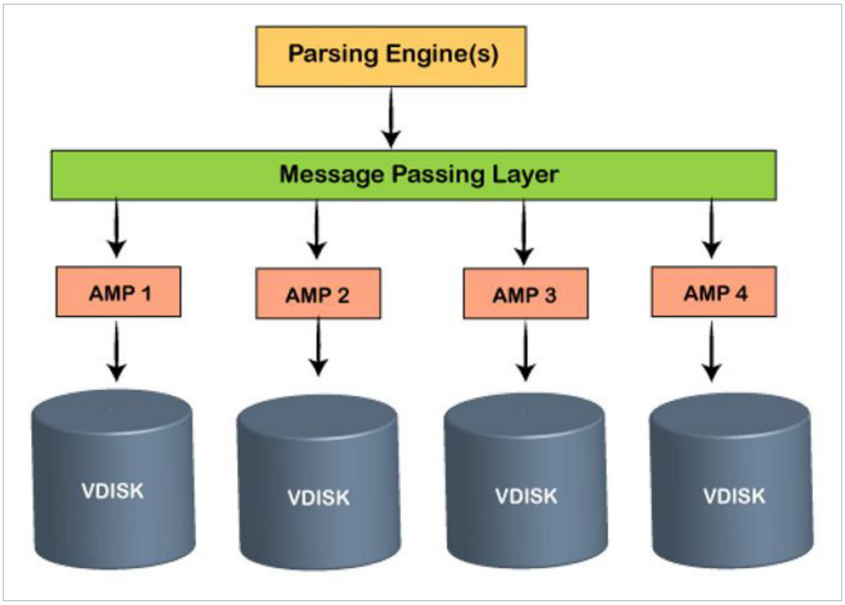

Figure 1: Teradata architecture

the access module processor (AMP) are responsible for storing and retrieving data.

The different components of a parsing engine are listed below.

- *Parser:* This decomposes SQL queries into relational data management processing steps.
- *Query optimizer:* This determines the most efficient path to access data.
- *Step generator:* This produces processing steps and encapsulates them into packages.
- *Dispatcher:* This transmits the encapsulated steps from the parser to the relevant AMPs, and performs monitoring and error-handling functionalities during step processing.
- *Session controller:* This manipulates session-control activities (e.g., log on, log off and authentication) and restores a session only after client or server failures.

Message passing layer: This is also called BYNET (Banyan network), and is the networking layer in the Teradata system. It allows interaction between PE and AMP, and also between the nodes. It receives the execution plan from the PE and sends it to AMP. Similarly, it receives the outputs from the AMP and sends them to PE. In the Teradata system, there are always two BYNET systems. These are called BYNET 0 and BYNET 1. But this is generally referred to as a single BYNET system. If one BYNET fails, the second one takes its place. But when data is large, both BYNETs can be made functional, which enhances the communication between PE and AMPs.

Disk: This component organises and stores the data while performing data manipulation activities. Teradata database employs redundant array of independent disks (RAID) storage technology to provide data security at the disk level. When AMPs are associated with disks, it is called Vdisks (virtual disks).

> **Note:** Each AMP is allowed to read and write on its own disk only. That is why it is also called shared-nothing architecture.

Teradata acts as a single data store that receives a large number of concurrent requests from various client applications. It is defined as an information repository supported by tools and utilities that make it, as part of Teradata Warehouse, a complete and active RDBMS.

Types of spaces in Teradata

Permanent space: This is the maximum amount of space allocated to the user/database to hold data rows. It is used for database objects (like permanent tables, indexes, etc) creation, and to hold their data. The total permanent space is divided among the total number of AMPs. Whenever the per AMP limit exceeds the allocated space of AMP, a 'No more room in database' error is generated. Permanent tables, journals, fallback tables, and secondary index sub-tables use permanent space. All databases have a predefined upper limit of permanent space. Teradata doesn't physically preallocate permanent space for databases and users when they are defined during object definition time.

Spool space: This is the unused permanent space that is used by the system to keep the intermediate results of the SQL query. Once the query is complete, the spool space is released. The volatile tables use the spool space. Users without spool space can't execute any query. Data is active up to the current session only. It is divided among the number of AMPs. Whenever the per AMP limit exceeds the allocated space, the user will get a spool space error.

Temporary space: This is the unused permanent space used by global temporary tables (GTT). It is divided by the number of AMPs. Data is active up to the current session only. It is reserved prior to spool space for any user defined operations. Temp space is allocated at the database level or user level, but not at the table level. This is the amount of space used for GTT, and these results